

PRECISION NDE SERVICES INC.

FINAL INSPECTION REPORT

Ultrasonic Thickness Survey

Report Number: PNDE-RPT-20241025-031-R0

Revision: 0 (Original Issue)

Date: November 14, 2024

Client: Great Lakes Chemical Processing (GLCP)

Location: Hamilton Plant, Ontario

Unit: Amine Treating Unit (ATU-100)

Work Order: WO-20241025-031

Scope: 150 Condition Monitoring Locations (CMLs)
18 piping lines

Survey Dates: November 4–7, 2024

Role	Name	Signature	Date
Prepared By:	Marcus Chen, UT Level II	_____	Nov 12, 2024
Reviewed By:	Sarah Thompson, UT Level II	_____	Nov 13, 2024
Approved By:	David Park, UT Level III	_____	Nov 14, 2024

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1.0 EXECUTIVE SUMMARY

Precision NDE Services Inc. (PNDE) performed an ultrasonic thickness (UT) survey of the Amine Treating Unit (ATU-100) piping at the Great Lakes Chemical Processing (GLCP) Hamilton Plant, Ontario, from November 4 through November 7, 2024. The survey encompassed 150 Condition Monitoring Locations (CMLs) distributed across 18 piping lines in sour gas, treated gas, lean amine, rich amine, regenerator overhead, reflux, steam, and cooling water services.

Of the 150 CMLs in scope, 148 were successfully inspected and 2 were documented as inaccessible. The survey identified **7 CRITICAL findings** (readings below minimum required thickness), **3 Alert findings** (readings approaching t-min), and **23 Monitor conditions** (elevated corrosion rates exceeding 10 mpy). The remaining 115 CMLs were classified as Acceptable.

The most significant finding was at CML 4-RA-101-06 (4" Rich Amine line, 90° elbow near T-102 Regenerator) where a measured thickness of 0.142" was recorded — below the minimum required thickness of 0.150". GLCP Inspection Coordinator Jennifer Walsh was immediately notified on-site, and a Fitness-for-Service (FFS) evaluation per API 579 has been initiated. A second CRITICAL finding at CML 6-RA-103-11 (6" Rich Amine Header weldolet) measured 0.168" against a t-min of 0.180", indicating a localized thin area.

Rich amine and regenerator overhead services exhibited the highest corrosion rates, consistent with the expected degradation mechanisms for amine treating systems (acid gas loading, amine degradation products). The cooling water line 3-CW-101 exhibited an anomalous condition where three CMLs returned readings exceeding the nominal wall thickness, suggesting internal scale deposits rather than metal loss. Internal inspection during the next turnaround is recommended.

Category	Count
Total CMLs in Scope	150
CMLs Inspected	148
Inaccessible	2
CRITICAL (below t-min)	7
Alert (within 110% of t-min)	3
Monitor (rate > 10 mpy)	23
Acceptable	115

2.0 SCOPE OF WORK

The scope of work was defined by Client Request for Quotation GLCP-RFQ-2024-089, dated September 15, 2024. The survey covered the following:

Line Number	Service	Size (in)	Material	Schedule	CML Count
2-RF-102	Reflux	2	316SS	40S	7
2-RV-102	Regen Overhead	2	A106 Gr.B	80	6
2-ST-101	Reboiler Steam	2	A106 Gr.B	40	6
3-CW-101	Amine Cooler CW	3	A106 Gr.B	40	6
3-CW-102	Amine Cooler CW	3	A106 Gr.B	40	5
3-RF-101	Reflux	3	316SS	40S	9
3-RV-101	Regen Overhead	3	A106 Gr.B	80	8
4-LA-101	Lean Amine Supply	4	A106 Gr.B	40	9
4-LA-102	Lean Amine Supply	4	A106 Gr.B	40	8
4-LA-104	L/R Exchanger	4	A106 Gr.B	40	10
4-RA-101	Rich Amine	4	A106 Gr.B	80	12
4-RA-102	Rich Amine	4	A106 Gr.B	80	11
4-RA-104	L/R Exchanger	4	A106 Gr.B	80	8
6-LA-103	Lean Amine Header	6	A106 Gr.B	40	10
6-RA-103	Rich Amine Header	6	A106 Gr.B	80	14
6-SG-101	Sour Gas Inlet	6	A106 Gr.B	40	8
6-SG-102	Sour Gas Inlet	6	A106 Gr.B	40	6
8-TG-101	Treated Gas	8	A106 Gr.B	40	7
TOTAL					150

Survey Dates: November 4–7, 2024 (4 working days)

Previous Survey: October 15–17, 2019 (baseline data for trending)

3.0 REFERENCE DOCUMENTS

API 570 — Piping Inspection Code: In-Service Inspection, Rating, Repair, and Alteration of Piping Systems, 4th Edition, February 2016

API 574 — Inspection Practices for Piping System Components, 4th Edition, November 2016

ASME B31.3 — Process Piping, 2022 Edition

ASME Section V — Nondestructive Examination, Article 5 — Ultrasonic Examination Methods for Materials, 2023 Edition

API 579-1/ASME FFS-1 — Fitness-For-Service, 3rd Edition, June 2016

ASNT SNT-TC-1A — Personnel Qualification and Certification in Nondestructive Testing, 2020 Edition

GLCP-NDT-001 Rev 3 — Great Lakes Chemical Processing — Nondestructive Testing Requirements for Piping Systems, Revision 3, January 2023

GLCP-RFQ-2024-089 — Request for Quotation — ATU-100 Piping UT Survey, September 15, 2024

PNDE-PROC-UT-005 Rev 7 — Precision NDE Services — Ultrasonic Thickness Measurement Procedure, Revision 7, March 2024

PNDE-JSA-20241025-031 — Job Safety Analysis — ATU-100 UT Survey, October 25, 2024

4.0 PERSONNEL & QUALIFICATIONS

4.1 Inspection Personnel

Name	Role	Cert Level	Cert Number	Expiry	Gauge Assigned
Marcus Chen	Lead Inspector	UT Level II	PNDE-UT2-0847	Mar 15, 2026	38DL Plus S/N 2847193
Sarah Thompson	Inspector	UT Level II	PNDE-UT2-1203	Aug 22, 2025	38DL Plus S/N 2901456
David Park	Project Manager / Approver	UT Level III	PNDE-UT3-0215	Jan 10, 2027	N/A (office review)

4.2 Qualification Confirmation

All field inspectors hold a minimum UT Level II certification in accordance with ASNT SNT-TC-1A and employer written practice, as required by GLCP-NDT-001 §3.2. Both Marcus Chen and Sarah Thompson meet the minimum experience requirements of 1,680 hours of UT inspection experience. Certification records are appended in Appendix C.

4.3 Site Safety Orientation

All personnel completed the GLCP Hamilton Plant site-specific safety orientation on November 1, 2024, conducted by Mike Sullivan (GLCP Safety Coordinator). Daily tailgate safety meetings were conducted each morning prior to commencing work, in accordance with GLCP-NDT-001 §3.3 and Job Safety Analysis PNDE-JSA-20241025-031. H■S personal monitors were issued and tested daily. No safety incidents occurred during the survey period.

5.0 EQUIPMENT & CALIBRATION

5.1 Ultrasonic Thickness Gauges

Item	Description	Serial Number	Cal Cert No.	Cal Due Date
UT Gauge 1	Olympus 38DL Plus	2847193	PNDE-CAL-2024-0847	May 15, 2025
UT Gauge 2	Olympus 38DL Plus	2901456	PNDE-CAL-2024-1203	Jul 22, 2025

5.2 Transducers

Transducer	Type	Frequency	Diameter	Serial Number
Probe 1	D790-SM, Dual Element	5 MHz	0.375"	D790-2847-A
Probe 2	D790-SM, Dual Element	5 MHz	0.375"	D790-2901-A

5.3 Calibration Block

Calibration Standard: PNDE-CS-001, 4-step carbon steel block (0.100", 0.200", 0.400", 0.800"), NIST-traceable. Certificate Number: PNDE-CAL-BLK-2024-001, expiry June 30, 2025. Calibration certificate is appended in Appendix D.

5.4 Material Velocity Settings

Material	Velocity (in/ μ s)	Lines
A106 Gr.B (Carbon Steel)	0.2330	All lines except 3-RF-101, 2-RF-102
316 Stainless Steel	0.2260	3-RF-101, 2-RF-102 (Reflux)

5.5 Daily Calibration Verification Records

Daily calibration checks were performed each morning prior to commencing inspections and at the end of each shift, in accordance with GLCP-NDT-001 §4.3. All readings were within the required ± 0.002 " tolerance. Calibration verification photographs are included in Appendix B.

Date	Inspector	Gauge S/N	AM Check (0.200" step)	PM Check (0.200" step)	Status
Nov 4, 2024	M. Chen	2847193	0.200"	0.201"	PASS
Nov 4, 2024	S. Thompson	2901456	0.201"	0.200"	PASS
Nov 5, 2024	M. Chen	2847193	0.200"	0.201"	PASS
Nov 5, 2024	S. Thompson	2901456	0.201"	0.200"	PASS
Nov 6, 2024	M. Chen	2847193	0.200"	0.201"	PASS
Nov 6, 2024	S. Thompson	2901456	0.201"	0.200"	PASS
Nov 7, 2024	M. Chen	2847193	0.200"	0.201"	PASS
Nov 7, 2024	S. Thompson	2901456	0.201"	0.200"	PASS

6.0 INSPECTION RESULTS — DATA TABLES

The following tables present the thickness measurement results organized by piping line and CML location. Each CML was measured using a minimum of 4 readings at 90° intervals around the pipe circumference, per GLCP-NDT-001 §5.3. The minimum reading from the four-point scan is reported as the measured thickness. Historical data from the October 2019 survey is included for trending comparison.

6.1 Line 2-RF-102 — Reflux (2" 316SS, Sch 40S)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
2-RF-102-01	Elbow 45°	Elevated platform EL+25', At direction change	0.154	0.085	0.125	0.098	5.4	Acceptable
2-RF-102-02	Reducer	Upstream of control valve, Size transition	0.154	0.085	0.126	0.093	6.6	Alert
2-RF-102-03	Elbow 90°	Upstream of control valve, At direction change	0.154	0.085	0.108	0.079	5.8	CRITICAL
2-RF-102-04	Straight Run	Grade level, 6 o'clock position	0.154	0.085	0.114	0.066	9.6	CRITICAL
2-RF-102-05	Straight Run	Near T-102 Regenerator, Near support	0.154	0.085	0.113	0.079	6.8	CRITICAL
2-RF-102-06	Reducer	Near V-101 Contactor, Size transition	0.154	0.085	0.132	0.121	2.2	Acceptable
2-RF-102-07	Straight Run	Near E-104 Amine Cooler, 6 o'clock position	0.154	0.085	0.115	0.098	3.4	Acceptable

6.2 Line 2-RV-102 — Regen Overhead (2" A106 Gr.B, Sch 80)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
2-RV-102-01	Tee	Near E-104 Amine Cooler, Branch connection	0.218	0.11	0.188	0.135	10.6	Monitor
2-RV-102-02	Straight Run	South side of pipe rack, 6 o'clock position	0.218	0.11	0.142	0.087	11	CRITICAL
2-RV-102-03	Reducer	Downstream of control valve, Size transition	0.218	0.11	0.179	0.149	6	Acceptable
2-RV-102-04	Flange	Near E-103 L/R Exchanger, At flange neck	0.218	0.11	0.138	0.098	8	CRITICAL
2-RV-102-05	Reducer	Near P-105A/B Amine Pumps, Size transition	0.218	0.11	0.149	0.119	6	Alert
2-RV-102-06	Elbow 90°	At PSV inlet, At direction change	0.218	0.11	0.147	0.126	4.2	Acceptable

6.3 Line 2-ST-101 — Reboiler Steam (2" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
2-ST-101-01	Straight Run	Upstream of control valve, Near support	0.154	0.09	0.133	0.111	4.4	Acceptable
2-ST-101-02	Elbow 45°	Grade level, At direction change	0.154	0.09	0.13	0.116	2.8	Acceptable

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
2-ST-101-03	Straight Run	Elevated platform EL+15', Near support	0.154	0.09	0.148	—	—	Inaccessible
2-ST-101-04	Tee	Near V-101 Contactor, Branch connection	0.154	0.09	0.144	0.132	2.4	Acceptable
2-ST-101-05	Straight Run	Near T-102 Regenerator, Pipe run	0.154	0.09	0.149	0.142	1.4	Acceptable
2-ST-101-06	Straight Run	Near V-101 Contactor, Near support	0.154	0.09	0.136	0.121	3	Acceptable

6.4 Line 3-CW-101 — Amine Cooler CW (3" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
3-CW-101-01	Flange	Near V-101 Contactor, At flange neck	0.216	0.11	0.193	0.228	-7	Acceptable
3-CW-101-02	Reducer	Elevated platform EL+25', Size transition	0.216	0.11	0.196	0.231	-7	Acceptable
3-CW-101-03	Straight Run	Near V-101 Contactor, 6 o'clock position	0.216	0.11	0.198	0.173	5	Acceptable
3-CW-101-04	Weldolet	Elevated platform EL+15', Branch outlet	0.216	0.11	0.202	0.225	-4.6	Acceptable
3-CW-101-05	Straight Run	Grade level, Pipe run	0.216	0.11	0.206	0.195	2.2	Acceptable
3-CW-101-06	Weldolet	Downstream of control valve, Branch outlet	0.216	0.11	0.195	0.175	4	Acceptable

6.5 Line 3-CW-102 — Amine Cooler CW (3" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
3-CW-102-01	Straight Run	Near P-105A/B Amine Pumps, Pipe run	0.216	0.11	0.199	0.182	3.4	Acceptable
3-CW-102-02	Elbow 45°	Upstream of control valve, At direction change	0.216	0.11	0.204	0.195	1.8	Acceptable
3-CW-102-03	Elbow 90°	Near P-105A/B Amine Pumps, At direction change	0.216	0.11	0.198	0.174	4.8	Acceptable
3-CW-102-04	Weldolet	Grade level, Branch outlet	0.216	0.11	0.195	0.173	4.4	Acceptable
3-CW-102-05	Flange	South side of pipe rack, At flange neck	0.216	0.11	0.193	0.179	2.8	Acceptable

6.6 Line 3-RF-101 — Reflux (3" 316SS, Sch 40S)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
3-RF-101-01	Straight Run	West header, Near support	0.216	0.095	0.197	0.166	6.2	Acceptable
3-RF-101-02	Elbow 90°	Near T-102 Regenerator, At direction change	0.216	0.095	0.194	0.166	5.6	Acceptable
3-RF-101-03	Flange	Near E-104 Amine Cooler, At flange neck	0.216	0.095	0.183	0.16	4.6	Acceptable
3-RF-101-04	Flange	Near E-103 L/R Exchanger, At flange neck	0.216	0.095	0.174	0.147	5.4	Acceptable

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
3-RF-101-05	Weldolet	Near V-101 Contactor, Branch outlet	0.216	0.095	0.17	0.141	5.8	Acceptable
3-RF-101-06	Straight Run	Downstream of control valve, Pipe run	0.216	0.095	0.194	0.153	8.2	Acceptable
3-RF-101-07	Weldolet	Near V-101 Contactor, Branch outlet	0.216	0.095	0.178	0.165	2.6	Acceptable
3-RF-101-08	Straight Run	West header, Pipe run	0.216	0.095	0.191	0.159	6.4	Acceptable
3-RF-101-09	Tee	East header, Branch connection	0.216	0.095	0.194	0.156	7.6	Acceptable

6.7 Line 3-RV-101 — Regen Overhead (3" A106 Gr.B, Sch 80)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
3-RV-101-01	Straight Run	North side of pipe rack, Near support	0.3	0.14	0.244	0.188	11.2	Monitor
3-RV-101-02	Reducer	Elevated platform EL+15', Size transition	0.3	0.14	0.25	0.18	14	Monitor
3-RV-101-03	Straight Run	Near E-104 Amine Cooler, 6 o'clock position	0.3	0.14	0.222	0.189	6.6	Acceptable
3-RV-101-04	Straight Run	Near E-104 Amine Cooler, Pipe run	0.3	0.14	0.249	0.195	10.8	Monitor
3-RV-101-05	Straight Run	Near T-102 Regenerator, Pipe run	0.3	0.14	0.267	0.196	14.2	Monitor
3-RV-101-06	Tee	Near E-103 L/R Exchanger, Branch connection	0.3	0.14	0.239	0.174	13	Monitor
3-RV-101-07	Straight Run	Grade level, Pipe run	0.3	0.14	0.263	0.212	10.2	Monitor
3-RV-101-08	Straight Run	Near V-101 Contactor, 6 o'clock position	0.3	0.14	0.248	0.202	9.2	Acceptable

6.8 Line 4-LA-101 — Lean Amine Supply (4" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
4-LA-101-01	Tee	Near E-104 Amine Cooler, Branch connection	0.237	0.125	0.218	0.184	6.8	Acceptable
4-LA-101-02	Straight Run	Elevated platform EL+25', Near support	0.237	0.125	0.201	0.181	4	Acceptable
4-LA-101-03	Weldolet	Upstream of control valve, Branch outlet	0.237	0.125	0.189	0.174	3	Acceptable
4-LA-101-04	Weldolet	Downstream of control valve, Branch outlet	0.237	0.125	0.214	0.2	2.8	Acceptable
4-LA-101-05	Straight Run	East header, Near support	0.237	0.125	0.208	0.163	9	Acceptable
4-LA-101-06	Straight Run	Near T-102 Regenerator, Pipe run	0.237	0.125	0.195	0.181	2.8	Acceptable
4-LA-101-07	Straight Run	Upstream of control valve, Pipe run	0.237	0.125	0.2	0.184	3.2	Acceptable
4-LA-101-08	Tee	Near T-102 Regenerator, Branch connection	0.237	0.125	0.19	0.143	9.4	Acceptable
4-LA-101-09	Straight Run	Near V-101 Contactor, 6 o'clock position	0.237	0.125	0.224	0.179	9	Acceptable

6.9 Line 4-LA-102 — Lean Amine Supply (4" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
4-LA-102-01	Straight Run	West header, Pipe run	0.237	0.125	0.222	0.174	9.6	Acceptable
4-LA-102-02	Elbow 45°	Near P-105A/B Amine Pumps, At direction change	0.237	0.125	0.204	0.164	8	Acceptable
4-LA-102-03	Tee	East header, Branch connection	0.237	0.125	0.201	0.184	3.4	Acceptable
4-LA-102-04	Weldolet	Upstream of control valve, Branch outlet	0.237	0.125	0.203	0.156	9.4	Acceptable
4-LA-102-05	Reducer	Near P-105A/B Amine Pumps, Size transition	0.237	0.125	0.214	0.178	7.2	Acceptable
4-LA-102-06	Flange	At PSV inlet, At flange neck	0.237	0.125	0.203	0.184	3.8	Acceptable
4-LA-102-07	Straight Run	North side of pipe rack, 6 o'clock position	0.237	0.125	0.201	0.182	3.8	Acceptable
4-LA-102-08	Elbow 45°	Near P-105A/B Amine Pumps, At direction change	0.237	0.125	0.199	0.153	9.2	Acceptable

6.10 Line 4-LA-104 — L/R Exchanger (4" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
4-LA-104-01	Flange	At PSV inlet, At flange neck	0.237	0.125	0.219	0.208	2.2	Acceptable
4-LA-104-02	Straight Run	Near E-103 L/R Exchanger, Near support	0.237	0.125	0.201	0.165	7.2	Acceptable
4-LA-104-03	Tee	Upstream of control valve, Branch connection	0.237	0.125	0.204	0.179	5	Acceptable
4-LA-104-04	Reducer	Near V-101 Contactor, Size transition	0.237	0.125	0.212	0.187	5	Acceptable
4-LA-104-05	Straight Run	Grade level, Near support	0.237	0.125	0.192	0.152	8	Acceptable
4-LA-104-06	Elbow 90°	Elevated platform EL+15', At direction change	0.237	0.125	0.213	0.165	9.6	Acceptable
4-LA-104-07	Elbow 90°	Downstream of control valve, At direction change	0.237	0.125	0.221	0.177	8.8	Acceptable
4-LA-104-08	Tee	West header, Branch connection	0.237	0.125	0.219	0.186	6.6	Acceptable
4-LA-104-09	Reducer	Near V-101 Contactor, Size transition	0.237	0.125	0.199	0.182	3.4	Acceptable
4-LA-104-10	Elbow 45°	Downstream of control valve, At direction change	0.237	0.125	0.195	0.181	2.8	Acceptable

6.11 Line 4-RA-101 — Rich Amine (4" A106 Gr.B, Sch 80)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
4-RA-101-01	Straight Run	North side of pipe rack, Near support	0.337	0.15	0.249	0.187	12.4	Monitor
4-RA-101-02	Flange	Grade level, At flange neck	0.337	0.15	0.287	0.212	15	Monitor
4-RA-101-03	Weldolet	Near E-104 Amine Cooler, Branch outlet	0.337	0.15	0.265	0.232	6.6	Acceptable
4-RA-101-04	Elbow 90°	Near E-103 L/R Exchanger, At direction change	0.337	0.15	0.264	0.23	6.8	Acceptable

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
4-RA-101-05	Elbow 90°	Near E-103 L/R Exchanger, At direction change	0.337	0.15	0.284	0.237	9.4	Acceptable
4-RA-101-06	Elbow 90°	Near T-102 Regenerator, At direction change	0.337	0.15	0.306	0.142	32.8	CRITICAL
4-RA-101-07	Elbow 45°	Near E-104 Amine Cooler, At direction change	0.337	0.15	0.257	0.186	14.2	Monitor
4-RA-101-08	Reducer	Near E-103 L/R Exchanger, Size transition	0.337	0.15	0.275	0.235	8	Acceptable
4-RA-101-09	Weldolet	Elevated platform EL+15', Branch outlet	0.337	0.15	0.252	0.194	11.6	Monitor
4-RA-101-10	Reducer	Grade level, Size transition	0.337	0.15	0.288	0.255	6.6	Acceptable
4-RA-101-11	Straight Run	At PSV inlet, 6 o'clock position	0.337	0.15	0.263	0.221	8.4	Acceptable
4-RA-101-12	Elbow 45°	South side of pipe rack, At direction change	0.337	0.15	0.278	0.255	4.6	Acceptable

6.12 Line 4-RA-102 — Rich Amine (4" A106 Gr.B, Sch 80)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
4-RA-102-01	Straight Run	At PSV inlet, 6 o'clock position	0.337	0.15	0.307	0.275	6.4	Acceptable
4-RA-102-02	Flange	Upstream of control valve, At flange neck	0.337	0.15	0.277	0.203	14.8	Monitor
4-RA-102-03	Weldolet	North side of pipe rack, Branch outlet	0.337	0.15	0.27	0.199	14.2	Monitor
4-RA-102-04	Elbow 45°	At PSV inlet, At direction change	0.337	0.15	0.302	0.281	4.2	Acceptable
4-RA-102-05	Weldolet	East header, Branch outlet	0.337	0.15	0.296	0.238	11.6	Monitor
4-RA-102-06	Reducer	Near E-104 Amine Cooler, Size transition	0.337	0.15	0.271	0.227	8.8	Acceptable
4-RA-102-07	Elbow 90°	South side of pipe rack, At direction change	0.337	0.15	0.273	0.255	3.6	Acceptable
4-RA-102-08	Straight Run	Downstream of control valve, 6 o'clock position	0.337	0.15	0.306	0.281	5	Acceptable
4-RA-102-09	Flange	Near T-102 Regenerator, At flange neck	0.337	0.15	0.268	0.161	21.4	Alert
4-RA-102-10	Elbow 45°	South side of pipe rack, At direction change	0.337	0.15	0.253	0.229	4.8	Acceptable
4-RA-102-11	Elbow 90°	Upstream of control valve, At direction change	0.337	0.15	0.289	0.248	8.2	Acceptable

6.13 Line 4-RA-104 — L/R Exchanger (4" A106 Gr.B, Sch 80)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
4-RA-104-01	Elbow 90°	At PSV inlet, At direction change	0.337	0.15	0.293	0.221	14.4	Monitor
4-RA-104-02	Straight Run	Near P-105A/B Amine Pumps, Near support	0.337	0.15	0.3	0.235	13	Monitor
4-RA-104-03	Reducer	South side of pipe rack, Size transition	0.337	0.15	0.267	0.211	11.2	Monitor

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
4-RA-104-04	Elbow 45°	Downstream of control valve, At direction change	0.337	0.15	0.251	0.215	7.2	Acceptable
4-RA-104-05	Straight Run	Near E-104 Amine Cooler, Near support	0.337	0.15	0.289	0.254	7	Acceptable
4-RA-104-06	Straight Run	Upstream of control valve, Near support	0.337	0.15	0.281	0.244	7.4	Acceptable
4-RA-104-07	Flange	Near P-105A/B Amine Pumps, At flange neck	0.337	0.15	0.301	0.255	9.2	Acceptable
4-RA-104-08	Straight Run	Near E-103 L/R Exchanger, Pipe run	0.337	0.15	0.261	0.217	8.8	Acceptable

6.14 Line 6-LA-103 — Lean Amine Header (6" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
6-LA-103-01	Straight Run	South side of pipe rack, 6 o'clock position	0.28	0.14	0.246	0.217	5.8	Acceptable
6-LA-103-02	Weldolet	East header, Branch outlet	0.28	0.14	0.241	0.208	6.6	Acceptable
6-LA-103-03	Reducer	Downstream of control valve, Size transition	0.28	0.14	0.248	0.234	2.8	Acceptable
6-LA-103-04	Straight Run	Elevated platform EL+15', Near support	0.28	0.14	0.25	0.211	7.8	Acceptable
6-LA-103-05	Elbow 45°	North side of pipe rack, At direction change	0.28	0.14	0.251	0.207	8.8	Acceptable
6-LA-103-06	Reducer	West header, Size transition	0.28	0.14	0.259	0.232	5.4	Acceptable
6-LA-103-07	Straight Run	Near V-101 Contactor, Pipe run	0.28	0.14	0.259	0.231	5.6	Acceptable
6-LA-103-08	Weldolet	South side of pipe rack, Branch outlet	0.28	0.14	0.266	—	—	Inaccessible
6-LA-103-09	Elbow 90°	South side of pipe rack, At direction change	0.28	0.14	0.234	0.217	3.4	Acceptable
6-LA-103-10	Flange	Near V-101 Contactor, At flange neck	0.28	0.14	0.258	0.246	2.4	Acceptable

6.15 Line 6-RA-103 — Rich Amine Header (6" A106 Gr.B, Sch 80)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
6-RA-103-01	Elbow 90°	Grade level, At direction change	0.432	0.18	0.386	0.313	14.6	Monitor
6-RA-103-02	Elbow 90°	At PSV inlet, At direction change	0.432	0.18	0.385	0.33	11	Monitor
6-RA-103-03	Straight Run	Downstream of control valve, 6 o'clock position	0.432	0.18	0.373	0.3	14.6	Monitor
6-RA-103-04	Tee	At PSV inlet, Branch connection	0.432	0.18	0.35	0.276	14.8	Monitor
6-RA-103-05	Tee	At PSV inlet, Branch connection	0.432	0.18	0.38	0.31	14	Monitor
6-RA-103-06	Straight Run	Downstream of control valve, Near support	0.432	0.18	0.366	0.336	6	Acceptable
6-RA-103-07	Tee	Downstream of control valve, Branch connection	0.432	0.18	0.362	0.345	3.4	Acceptable
6-RA-103-08	Flange	West header, At flange neck	0.432	0.18	0.396	0.36	7.2	Acceptable
6-RA-103-09	Reducer	West header, Size transition	0.432	0.18	0.405	0.37	7	Acceptable

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
6-RA-103-10	Elbow 45°	Downstream of control valve, At direction change	0.432	0.18	0.39	0.351	7.8	Acceptable
6-RA-103-11	Weldolet	Elevated platform EL+15', Branch outlet	0.432	0.18	0.346	0.168	35.6	CRITICAL
6-RA-103-12	Straight Run	East header, Pipe run	0.432	0.18	0.372	0.355	3.4	Acceptable
6-RA-103-13	Straight Run	Elevated platform EL+15', Near support	0.432	0.18	0.36	0.309	10.2	Monitor
6-RA-103-14	Weldolet	South side of pipe rack, Branch outlet	0.432	0.18	0.383	0.353	6	Acceptable

6.16 Line 6-SG-101 — Sour Gas Inlet (6" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
6-SG-101-01	Elbow 45°	North side of pipe rack, At direction change	0.28	0.14	0.238	0.222	3.2	Acceptable
6-SG-101-02	Elbow 45°	Near E-104 Amine Cooler, At direction change	0.28	0.14	0.239	0.205	6.8	Acceptable
6-SG-101-03	Elbow 90°	North side of pipe rack, At direction change	0.28	0.14	0.262	0.228	6.8	Acceptable
6-SG-101-04	Weldolet	West header, Branch outlet	0.28	0.14	0.241	0.214	5.4	Acceptable
6-SG-101-05	Flange	Near E-103 L/R Exchanger, At flange neck	0.28	0.14	0.257	0.217	8	Acceptable
6-SG-101-06	Tee	Near P-105A/B Amine Pumps, Branch connection	0.28	0.14	0.25	0.231	3.8	Acceptable
6-SG-101-07	Straight Run	South side of pipe rack, 6 o'clock position	0.28	0.14	0.261	0.217	8.8	Acceptable
6-SG-101-08	Straight Run	Downstream of control valve, Pipe run	0.28	0.14	0.264	0.249	3	Acceptable

6.17 Line 6-SG-102 — Sour Gas Inlet (6" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
6-SG-102-01	Straight Run	South side of pipe rack, Near support	0.28	0.14	0.247	0.212	7	Acceptable
6-SG-102-02	Straight Run	Near E-103 L/R Exchanger, 6 o'clock position	0.28	0.14	0.257	0.221	7.2	Acceptable
6-SG-102-03	Straight Run	South side of pipe rack, Pipe run	0.28	0.14	0.236	0.211	5	Acceptable
6-SG-102-04	Flange	Near E-104 Amine Cooler, At flange neck	0.28	0.14	0.234	0.21	4.8	Acceptable
6-SG-102-05	Straight Run	Near P-105A/B Amine Pumps, Pipe run	0.28	0.14	0.232	0.198	6.8	Acceptable
6-SG-102-06	Tee	Near T-102 Regenerator, Branch connection	0.28	0.14	0.238	0.213	5	Acceptable

6.18 Line 8-TG-101 — Treated Gas (8" A106 Gr.B, Sch 40)

CML ID	Component	Location	Nom. (in)	t-min (in)	2019 (in)	2024 (in)	Rate (mpy)	Status
8-TG-101-01	Weldolet	West header, Branch outlet	0.322	0.16	0.304	0.283	4.2	Acceptable
8-TG-101-02	Reducer	Upstream of control valve, Size transition	0.322	0.16	0.314	0.304	2	Acceptable
8-TG-101-03	Reducer	At PSV inlet, Size transition	0.322	0.16	0.299	0.288	2.2	Acceptable
8-TG-101-04	Flange	Elevated platform EL+25', At flange neck	0.322	0.16	0.299	0.291	1.6	Acceptable
8-TG-101-05	Tee	West header, Branch connection	0.322	0.16	0.302	0.282	4	Acceptable
8-TG-101-06	Straight Run	At PSV inlet, Near support	0.322	0.16	0.304	0.279	5	Acceptable
8-TG-101-07	Tee	Near T-102 Regenerator, Branch connection	0.322	0.16	0.307	0.285	4.4	Acceptable

7.0 TRENDING & CORROSION ANALYSIS

7.1 Trending Comparison with Historical Data

Thickness readings from the current 2024 survey were compared against the baseline 2019 survey data to establish corrosion trends across all inspected CMLs. The interval between surveys is approximately 5.05 years (October 2019 to November 2024).

7.2 Corrosion Rate Summary by Line

Line	Service	Min Rate (mpy)	Max Rate (mpy)	Avg Rate (mpy)	Concern Level
2-RF-102	Reflux	2.2	9.6	5.7	Medium
2-RV-102	Regen Overhead	4.2	11.0	7.6	Medium
2-ST-101	Reboiler Steam	1.4	4.4	2.8	Low
3-CW-101	Amine Cooler CW	-7.0	5.0	-1.2	Low
3-CW-102	Amine Cooler CW	1.8	4.8	3.4	Low
3-RF-101	Reflux	2.6	8.2	5.8	Medium
3-RV-101	Regen Overhead	6.6	14.2	11.2	High
4-LA-101	Lean Amine Supply	2.8	9.4	5.6	Medium
4-LA-102	Lean Amine Supply	3.4	9.6	6.8	Medium
4-LA-104	L/R Exchanger	2.2	9.6	5.9	Medium
4-RA-101	Rich Amine	4.6	32.8	11.4	High
4-RA-102	Rich Amine	3.6	21.4	9.4	Medium
4-RA-104	L/R Exchanger	7.0	14.4	9.8	Medium
6-LA-103	Lean Amine Header	2.4	8.8	5.4	Medium
6-RA-103	Rich Amine Header	3.4	35.6	11.1	High
6-SG-101	Sour Gas Inlet	3.0	8.8	5.7	Medium
6-SG-102	Sour Gas Inlet	4.8	7.2	6.0	Medium
8-TG-101	Treated Gas	1.6	5.0	3.3	Low

7.3 Remaining Life and Next Inspection Date

Remaining life is calculated per API 570 §7.1 as: $RL = (t_{\text{actual}} - t_{\text{min}}) / \text{corrosion_rate}$. The next recommended inspection date is set at half the calculated remaining life (half-life method), capped at a maximum of 10 years, per API 570 §6.

CML ID	Line	2024 (in)	t-min (in)	Rate (mpy)	Remaining Life (yr)	Next Inspection
2-RF-102-02	2-RF-102	0.093	0.085	6.6	1.2	Jun 2025
2-RF-102-03	2-RF-102	0.079	0.085	5.8	-1.0	Immediate
2-RF-102-04	2-RF-102	0.066	0.085	9.6	-2.0	Immediate
2-RF-102-05	2-RF-102	0.079	0.085	6.8	-0.9	Immediate
2-RV-102-01	2-RV-102	0.135	0.11	10.6	2.4	Jan 2026
2-RV-102-02	2-RV-102	0.087	0.11	11	-2.1	Immediate

CML ID	Line	2024 (in)	t-min (in)	Rate (mpy)	Remaining Life (yr)	Next Inspection
2-RV-102-04	2-RV-102	0.098	0.11	8	-1.5	Immediate
2-RV-102-05	2-RV-102	0.119	0.11	6	1.5	Aug 2025
3-RV-101-01	3-RV-101	0.188	0.14	11.2	4.3	Jan 2027
3-RV-101-02	3-RV-101	0.18	0.14	14	2.9	Apr 2026
3-RV-101-04	3-RV-101	0.195	0.14	10.8	5.1	May 2027
3-RV-101-05	3-RV-101	0.196	0.14	14.2	3.9	Oct 2026
3-RV-101-06	3-RV-101	0.174	0.14	13	2.6	Feb 2026
3-RV-101-07	3-RV-101	0.212	0.14	10.2	7.1	May 2028
4-RA-101-01	4-RA-101	0.187	0.15	12.4	3.0	May 2026
4-RA-101-02	4-RA-101	0.212	0.15	15	4.1	Nov 2026
4-RA-101-06	4-RA-101	0.142	0.15	32.8	-0.2	Immediate
4-RA-101-07	4-RA-101	0.186	0.15	14.2	2.5	Feb 2026
4-RA-101-09	4-RA-101	0.194	0.15	11.6	3.8	Oct 2026
4-RA-102-02	4-RA-102	0.203	0.15	14.8	3.6	Aug 2026
4-RA-102-03	4-RA-102	0.199	0.15	14.2	3.5	Aug 2026
4-RA-102-05	4-RA-102	0.238	0.15	11.6	7.6	Aug 2028
4-RA-102-09	4-RA-102	0.161	0.15	21.4	0.5	May 2025
4-RA-104-01	4-RA-104	0.221	0.15	14.4	4.9	Apr 2027
4-RA-104-02	4-RA-104	0.235	0.15	13	6.5	Feb 2028
4-RA-104-03	4-RA-104	0.211	0.15	11.2	5.4	Jul 2027
6-RA-103-01	6-RA-103	0.313	0.18	14.6	9.1	May 2029
6-RA-103-02	6-RA-103	0.33	0.18	11	13.6	Aug 2031
6-RA-103-03	6-RA-103	0.3	0.18	14.6	8.2	Dec 2028
6-RA-103-04	6-RA-103	0.276	0.18	14.8	6.5	Feb 2028
6-RA-103-05	6-RA-103	0.31	0.18	14	9.3	Jul 2029
6-RA-103-11	6-RA-103	0.168	0.18	35.6	-0.3	Immediate
6-RA-103-13	6-RA-103	0.309	0.18	10.2	12.6	Feb 2031

8.0 FINDINGS & DISPOSITIONS

All findings are categorized per GLCP-NDT-001 §6.2 as follows:

Category	Criteria	Required Action
CRITICAL	Measured thickness below t-min	Immediate notification; engineering review; FFS evaluation
Alert	Measured thickness below 110% of t-min	1-year re-inspection interval; engineering notification
Monitor	Corrosion rate exceeding 10 mpy	Increased monitoring frequency; process review
Acceptable	Readings above thresholds, rates < 10 mpy	Standard re-inspection per API 570 §6

8.1 CRITICAL Findings — Readings Below t-min

Per GLCP-NDT-001 §6.3, all CRITICAL findings were reported to GLCP Inspection Coordinator Jennifer Walsh immediately upon discovery. The following CMLs recorded measured thickness values below the minimum required thickness:

CML ID	Line	Component	2024 (in)	t-min (in)	Rate (mpy)	Disposition
2-RF-102-03	2-RF-102	Elbow 90°	0.079	0.085	5.8	FFS evaluation per API 579; repair or replace as determined
2-RF-102-04	2-RF-102	Straight Run	0.066	0.085	9.6	FFS evaluation per API 579; repair or replace as determined
2-RF-102-05	2-RF-102	Straight Run	0.079	0.085	6.8	FFS evaluation per API 579; repair or replace as determined
2-RV-102-02	2-RV-102	Straight Run	0.087	0.11	11	FFS evaluation per API 579; repair or replace as determined
2-RV-102-04	2-RV-102	Flange	0.098	0.11	8	FFS evaluation per API 579; repair or replace as determined
4-RA-101-06	4-RA-101	Elbow 90°	0.142	0.15	32.8	Replace at next turnaround; FFS evaluation
6-RA-103-11	6-RA-103	Weldolet	0.168	0.18	35.6	FFS evaluation per API 579

Notification Documentation:

- **4-RA-101-06:** Reported to Jennifer Walsh (GLCP) on November 5, 2024 at 11:45 AM on-site. Reading confirmed by triple re-measurement. FFS evaluation initiated same day. Gauge display and surface condition photographs: IMG_0151, IMG_0202.
- **6-RA-103-11:** Reported to Jennifer Walsh (GLCP) on November 6, 2024 at 10:30 AM. Localized thin area mapped with circumferential readings. Gauge display and surface condition photographs: IMG_0163, IMG_0203.
- **2-RV-102-02, 2-RV-102-04:** Reported to Jennifer Walsh on November 6, 2024 at 2:00 PM. Multiple CMLs on this line below t-min. Line-wide replacement recommended. Photographs: IMG_0172, IMG_0173, IMG_0204, IMG_0205.
- **2-RF-102-03, 2-RF-102-04, 2-RF-102-05:** Reported to Jennifer Walsh on November 7, 2024 at 10:00 AM. 316SS reflux line showing general wall loss pattern. Photographs: IMG_0176, IMG_0177, IMG_0178, IMG_0206, IMG_0207, IMG_0208.

8.2 Alert Findings — Below 110% of t-min

CML ID	Line	Component	2024 (in)	110% t-min (in)	t-min (in)	Disposition
2-RF-102-02	2-RF-102	Reducer	0.093	0.094	0.085	1-year re-inspection interval
2-RV-102-05	2-RV-102	Reducer	0.119	0.121	0.11	1-year re-inspection interval
4-RA-102-09	4-RA-102	Flange	0.161	0.165	0.15	1-year re-inspection interval

8.3 Monitor Findings — Corrosion Rates Exceeding 10 mpy

CML ID	Line	Component	2024 (in)	Rate (mpy)	Disposition
4-RA-101-02	4-RA-101	Flange	0.212	15	2-year re-inspection; process review
4-RA-102-02	4-RA-102	Flange	0.203	14.8	2-year re-inspection; process review
6-RA-103-04	6-RA-103	Tee	0.276	14.8	2-year re-inspection; process review
6-RA-103-01	6-RA-103	Elbow 90°	0.313	14.6	2-year re-inspection; process review
6-RA-103-03	6-RA-103	Straight Run	0.3	14.6	2-year re-inspection; process review
4-RA-104-01	4-RA-104	Elbow 90°	0.221	14.4	2-year re-inspection; process review
4-RA-101-07	4-RA-101	Elbow 45°	0.186	14.2	2-year re-inspection; process review
4-RA-102-03	4-RA-102	Weldolet	0.199	14.2	2-year re-inspection; process review
3-RV-101-05	3-RV-101	Straight Run	0.196	14.2	2-year re-inspection; process review
6-RA-103-05	6-RA-103	Tee	0.31	14	2-year re-inspection; process review
3-RV-101-02	3-RV-101	Reducer	0.18	14	2-year re-inspection; process review
3-RV-101-06	3-RV-101	Tee	0.174	13	2-year re-inspection; process review
4-RA-104-02	4-RA-104	Straight Run	0.235	13	2-year re-inspection; process review
4-RA-101-01	4-RA-101	Straight Run	0.187	12.4	2-year re-inspection; process review
4-RA-101-09	4-RA-101	Weldolet	0.194	11.6	2-year re-inspection; process review
4-RA-102-05	4-RA-102	Weldolet	0.238	11.6	2-year re-inspection; process review
3-RV-101-01	3-RV-101	Straight Run	0.188	11.2	2-year re-inspection; process review
4-RA-104-03	4-RA-104	Reducer	0.211	11.2	2-year re-inspection; process review
6-RA-103-02	6-RA-103	Elbow 90°	0.33	11	2-year re-inspection; process review
3-RV-101-04	3-RV-101	Straight Run	0.195	10.8	2-year re-inspection; process review
2-RV-102-01	2-RV-102	Tee	0.135	10.6	2-year re-inspection; process review
6-RA-103-13	6-RA-103	Straight Run	0.309	10.2	2-year re-inspection; process review
3-RV-101-07	3-RV-101	Straight Run	0.212	10.2	2-year re-inspection; process review

8.4 Inaccessible CMLs

CML ID	Line	Component	Reason	Documentation
6-LA-103-08	6-LA-103	Weldolet	Scaffold could not reach. Pipe rack elevation. Defer to next turnaround.	Photo + field note
2-ST-101-03	2-ST-101	Straight Run	Insulated, client declined insulation removal. CUI inspection recommended.	Photo + field note

8.5 Anomalies

Scale Buildup — Line 3-CW-101 (Amine Cooler Cooling Water): CMLs 3-CW-101-01, 3-CW-101-02, and 3-CW-101-04 returned ultrasonic thickness readings exceeding the nominal wall thickness (readings of 0.228", 0.231", and 0.225" versus 0.216" nominal). This is indicative of internal scale or mineral deposits on the pipe inner surface, which increases the apparent wall thickness measured by the UT gauge. Gauge calibration was verified and confirmed accurate. Internal visual inspection is recommended during the next turnaround to assess scale extent and determine if cleaning is required.

Accelerated Corrosion — CML 3-RV-101-04: This CML on the Regenerator Overhead line exhibited a corrosion rate of approximately 12 mpy, significantly higher than the expected 5 mpy for this service. This may

indicate process upset conditions (e.g., acid gas carryover, amine degradation). Process engineering review is recommended.

9.0 RECOMMENDATIONS

9.1 Immediate Actions (Within 30 Days)

1. Complete Fitness-for-Service (FFS) evaluation per API 579-1 for CML 4-RA-101-06 (4" Rich Amine elbow, 0.142" vs 0.150" t-min). Engineering assessment to determine suitability for continued service, or requirement for repair/replacement.
2. Complete FFS evaluation for CML 6-RA-103-11 (6" Rich Amine Header weldolet, 0.168" vs 0.180" t-min). Map extent of localized thinning with supplemental grid scanning.
3. Evaluate 2" Regen Overhead line 2-RV-102 for potential near-term replacement. Three CMLs are below or near t-min (0.087", 0.098", 0.119" vs 0.110" t-min).
4. Evaluate 2" Reflux line 2-RF-102 for replacement. Three CMLs below t-min in 316SS service (0.066", 0.079", 0.079" vs 0.085" t-min).

9.2 Re-Inspection Intervals

Priority	Lines / CMLs	Interval	Basis
1 — Immediate	4-RA-101-06, 6-RA-103-11, 2-RV-102-02, 2-RV-102-04, 2-RF-102-03, 2-RF-102-04, 2-RF-102-05	FFS / Repair	Below t-min
2 — 1 Year	4-RA-102-09, 2-RV-102-05, 2-RF-102-02	Nov 2025	Within 110% of t-min
3 — 2 Years	All Monitor CMLs (23 locations) — Rich Amine, Regen OH lines	Nov 2026	Corrosion rate > 10 mpy
4 — 5 Years	All remaining Acceptable CMLs	Nov 2029	API 570 §6 half-life method

9.3 Engineering Review Referrals

The following items are referred to GLCP Process and Mechanical Engineering for review:

1. FFS evaluation per API 579-1, Part 4 (General Metal Loss) for CMLs 4-RA-101-06 and 6-RA-103-11. Part 5 (Local Metal Loss) assessment is recommended for 6-RA-103-11 given the localized nature of the thinning.
2. Metallurgical review of 2" Reflux line 2-RF-102 (316SS). The general wall loss pattern across multiple CMLs may indicate a material selection concern or chemical incompatibility.
3. Corrosion rate assessment for Rich Amine service. Rates of 12–35 mpy on Rich Amine lines suggest potential for amine degradation products (Heat Stable Salts) or acid gas loading beyond design parameters.

9.4 Process Review Recommendations

1. Review amine solution quality (HSS content, amine concentration, pH) for potential correlation with accelerated corrosion rates observed in Rich Amine and Regenerator Overhead services.
2. Investigate whether any process upsets occurred between the 2019 and 2024 survey periods that could explain the elevated corrosion at CML 3-RV-101-04 (12 mpy vs expected 5 mpy).
3. Schedule internal inspection of cooling water line 3-CW-101 during next turnaround to assess the extent of scale deposits and determine if chemical cleaning is warranted.

4. Consider installation of corrosion coupons or on-line corrosion monitoring probes on Rich Amine lines (4-RA-101, 4-RA-102, 6-RA-103) to provide real-time corrosion data between UT surveys.

APPENDIX A — RAW DATA (GAUGE DATA EXPORTS)

Raw gauge data was exported from both Olympus 38DL Plus units (S/N 2847193 and S/N 2901456) at the end of each survey day. Daily export files are archived as:

Filename	Records	Date	Lines Covered
field_data_20241104.csv	40 readings	Nov 4, 2024	6-SG-101, 6-SG-102, 4-LA-101, 4-LA-102
field_data_20241105.csv	40 readings	Nov 5, 2024	6-LA-103, 4-RA-101, 4-RA-102
field_data_20241106.csv	41 readings	Nov 6, 2024	4-RA-102, 6-RA-103, 3-RV-101, 2-RV-102
field_data_20241107.csv	29 readings	Nov 7, 2024	3-RF-101, 2-RF-102, 2-ST-101, 3-CW-101, 3-CW-102

A combined and reconciled data file (field_data_combined.csv) containing all 150 CML records with matched historical data is maintained by PNDE. Original gauge data files are retained per PNDE document control procedures for a minimum of 10 years.

APPENDIX B — PHOTOGRAPH LOG

A total of 226 photographs were taken during the survey, organized by the following categories per GLCP-NDT-001 §7.2:

Photo Type	Count	Description
CML Location	148	Every inspected CML — tag and surrounding area
Gauge Display	53	CRITICAL, Alert, and Monitor readings — gauge screen showing value
Surface Condition	7	Pipe surface at CRITICAL findings — visual assessment
Calibration Check	8	Daily AM calibration verification setup (2 inspectors × 4 days)
Inaccessible Documentation	2	Access obstruction — 6-LA-103-08 and 2-ST-101-03
Anomaly	3	Scale deposit evidence at 3-CW-101-01, -02, -04
General Site	5	Unit overview and work area orientation

All photographs are cross-referenced by CML ID in the photo manifest (photo_manifest.csv). High-resolution originals are archived in PNDE project files. Representative photographs for CRITICAL findings are referenced in Section 8.1 of this report.

Key Photographs — CRITICAL and Alert Findings:

CML ID	Status	Photo IDs	Photo Types
4-RA-101-06	CRITICAL	IMG_0053, IMG_0151, IMG_0202	CML location, gauge display, surface condition
6-RA-103-11	CRITICAL	IMG_0081, IMG_0163, IMG_0203	CML location, gauge display, surface condition
2-RV-102-02	CRITICAL	IMG_0094, IMG_0172, IMG_0204	CML location, gauge display, surface condition
2-RV-102-04	CRITICAL	IMG_0096, IMG_0173, IMG_0205	CML location, gauge display, surface condition
2-RF-102-03	CRITICAL	IMG_0110, IMG_0176, IMG_0206	CML location, gauge display, surface condition
2-RF-102-04	CRITICAL	IMG_0111, IMG_0177, IMG_0207	CML location, gauge display, surface condition
2-RF-102-05	CRITICAL	IMG_0112, IMG_0178, IMG_0208	CML location, gauge display, surface condition
4-RA-102-09	Alert	IMG_0068, IMG_0157	CML location, gauge display
2-RV-102-05	Alert	IMG_0097, IMG_0174	CML location, gauge display
2-RF-102-02	Alert	IMG_0109, IMG_0175	CML location, gauge display

APPENDIX C — PERSONNEL CERTIFICATION RECORDS

The following personnel certification records are maintained on file and are available upon request. Copies of current certifications were provided to GLCP prior to mobilization, in accordance with GLCP-NDT-001 §3.1.

Inspector	Cert Number	Method/Level	Issue Date	Expiry Date	Certifying Body	Document
Marcus Chen	PNDE-UT2-0847	UT Level II	Mar 15, 2021	Mar 15, 2026	PNDE per ASNT SNT-TC-1A	PersonnelCert_Marcus_Chen.pdf
Sarah Thompson	PNDE-UT2-1203	UT Level II	Aug 22, 2020	Aug 22, 2025	PNDE per ASNT SNT-TC-1A	PersonnelCert_Sarah_Thompson.pdf
David Park	PNDE-UT3-0215	UT Level III	Jan 10, 2022	Jan 10, 2027	PNDE per ASNT SNT-TC-1A	On file at PNDE

Note: Full certification packages including training records, examination results, and eye examination records are maintained at PNDE head office per ASNT SNT-TC-1A requirements.

APPENDIX D — EQUIPMENT CALIBRATION CERTIFICATES

The following calibration certificates are maintained on file. All calibrations are performed by PNDE's calibration laboratory, which maintains NIST traceability in accordance with GLCP-NDT-001 §4.2.

Equipment	Serial No.	Cal Cert No.	Cal Date	Due Date	Traceable To	Document
Olympus 38DL Plus	2847193	PNDE-CAL-2024-0847	May 15, 2024	May 15, 2025	NIST	CalCert_38DL-2847193.pdf
Olympus 38DL Plus	2901456	PNDE-CAL-2024-1203	Jul 22, 2024	Jul 22, 2025	NIST	CalCert_38DL-2901456.pdf
Cal Block (4-step CS)	PNDE-CS-001	PNDE-CAL-BLK-2024-001	Jun 30, 2024	Jun 30, 2025	NIST	CalCert_PNDE-CS-001.pdf

APPENDIX E — ISOMETRIC DRAWINGS WITH CML LOCATIONS

Isometric drawings showing CML locations are provided for the piping lines with CRITICAL findings. CML markers are indicated on the drawings with their identification numbers. CRITICAL and Alert CMLs are highlighted.

Drawing No.	Line	Service	CMLs Shown	CRITICAL CMLs
ISO_4-RA-101	4-RA-101	Rich Amine	12	4-RA-101-06
ISO_6-RA-103	6-RA-103	Rich Amine Header	14	6-RA-103-11

Note: Full-size isometric drawings (ISO_4-RA-101.svg, ISO_6-RA-103.svg) are provided as separate electronic files. Isometric drawings for all remaining lines are available from GLCP and can be marked up upon request.

— END OF REPORT —

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